

OpsBrain AI

Unified Asset & Operations Brain for Zero-Harm Plants

ET AI Hackathon 2026 | Problem Statement #8

Operational Silos & SCADA Blindspots

- **Sensor Blind Spots**

SCADA alarms warn of pressure spikes but do not know isolation valve status or the mandated safety steps.

- **The Knowledge Cliff**

Retiring senior engineers carry years of undocumented plant heuristics away, leaving junior operators with sparse guidance.

- **Audit Latency**

Cross-referencing safety compliance logs manually during an alert takes hours, slowing emergency response.

SCADA Telemetry

Out-of-context sensor readings

Regulatory SOPs

Trapped in static PDF binders

Asset Blueprints

Rasterized blueprints
disconnected from state

Incident Logs

Unstructured notebooks & logs

OpsBrain AI Architecture

- Physical Topology Mapping

Extracts connection rules and node properties directly from Piping & Instrumentation Diagrams (P&IDs).

- Local Vector Indexing

Ingests environmental and safety SOPs locally on CPU using BGE embeddings for absolute data privacy.

- Orchestration Engine

Deploys a fleet of LLM agents to monitor risks and offer prototype compliance evidence support.



Knowledge Graph Digital Twin

- Gemini Vision Parser

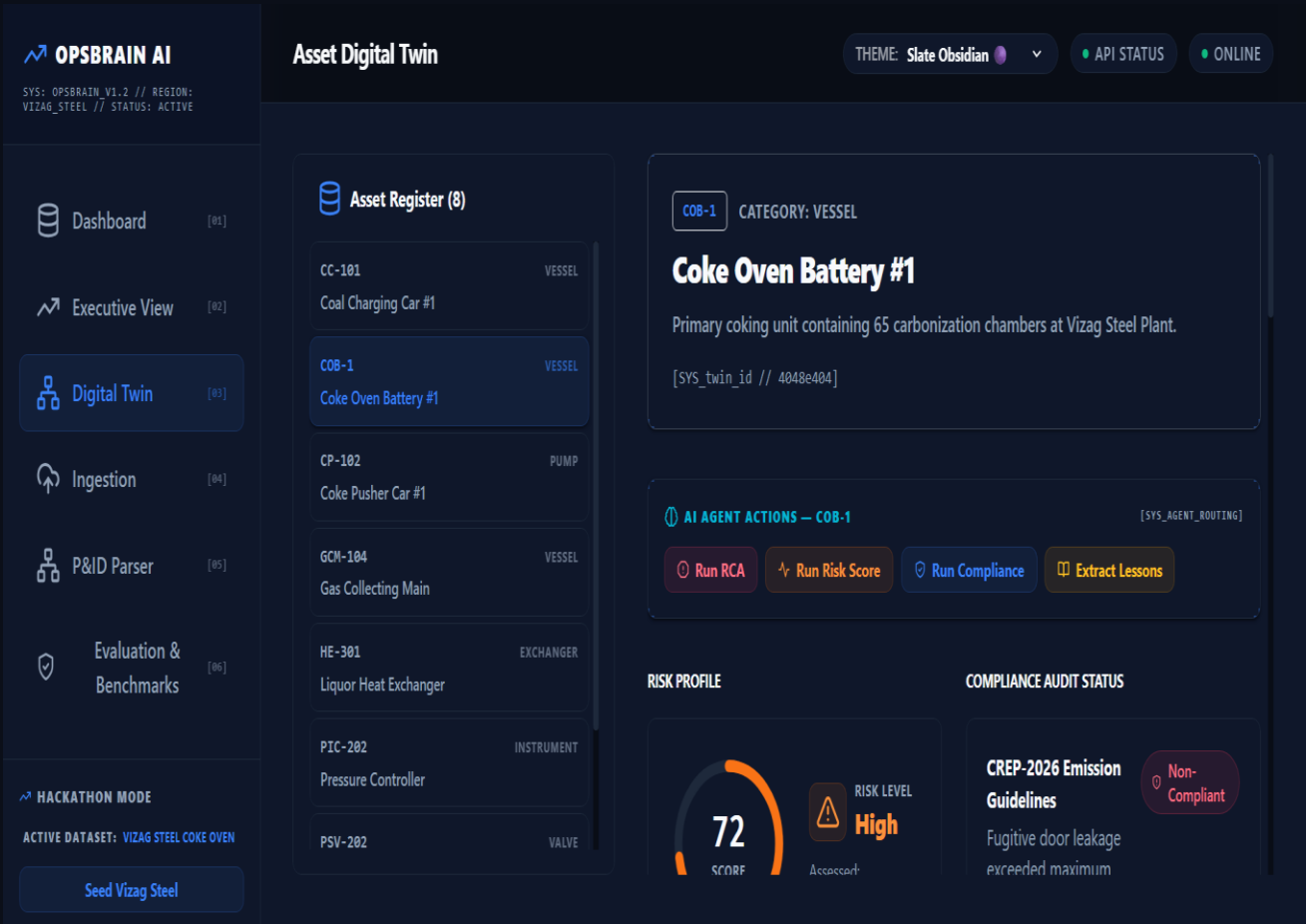
Vision model parses blueprint symbols, coordinates, flow directions, and equipment tags from drawings.

- Relational Graph

Renders equipment tags, flow lines, and sensor limits dynamically using ReactFlow.

- Local Security

Documents and asset links are stored on-premise, bypassing cloud risk vectors for heavy industries.



Specialized Multi-Agent Core

- Task-Focused Agents

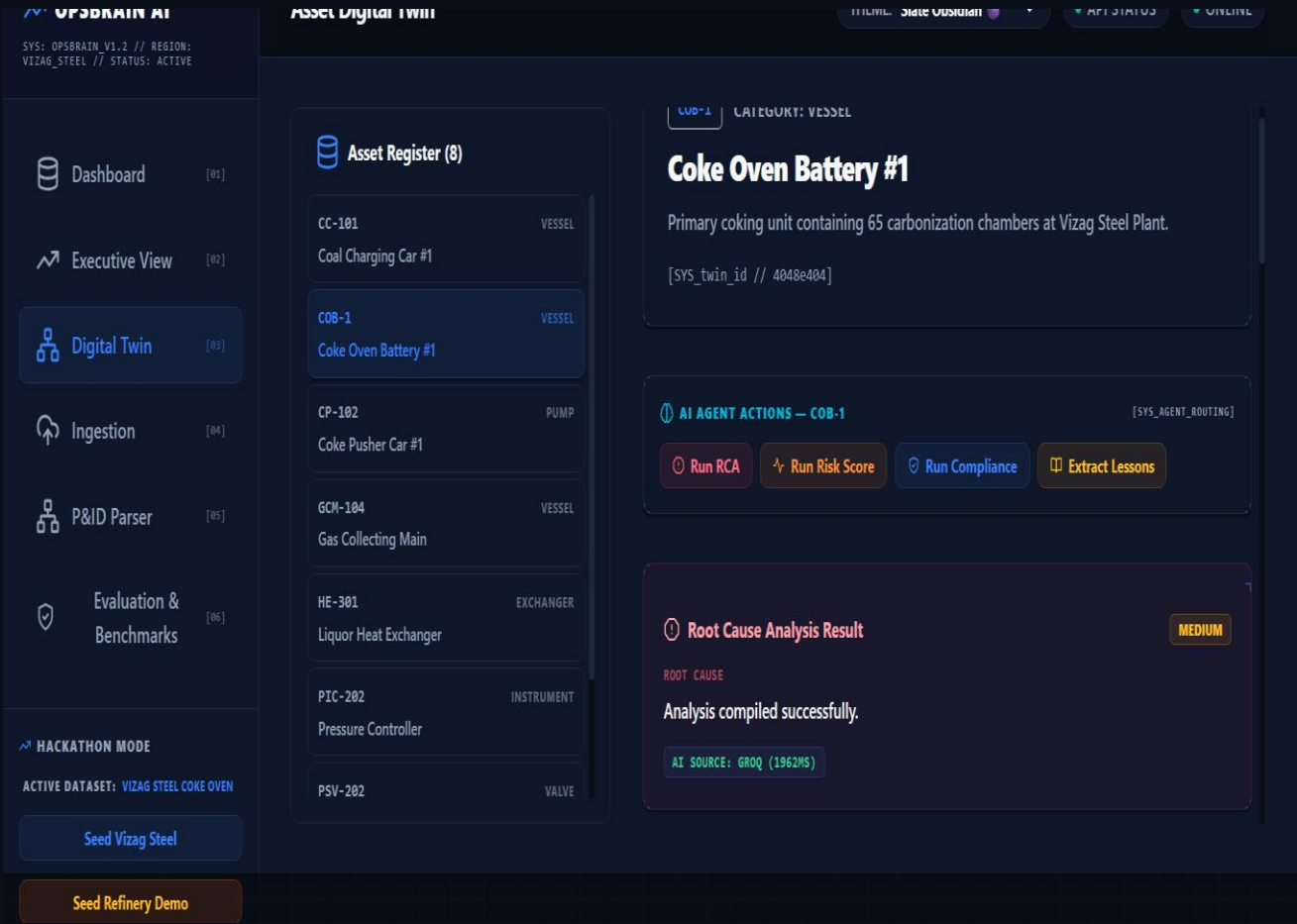
Five specialized agents (Root Cause, Risk, Compliance, Lessons, Copilot) handle targeted safety computations.

- AI Traversal HUD

Console overlay blocks screen during investigation to display detailed agent query and logic logs.

- Dynamic Graph Pulse

Pulsing warn highlights along traversed ReactFlow nodes track risk cascades on screen in real time.



Knowledge Copilot & Graph RAG

- **Graph-Aware Evidence Path**

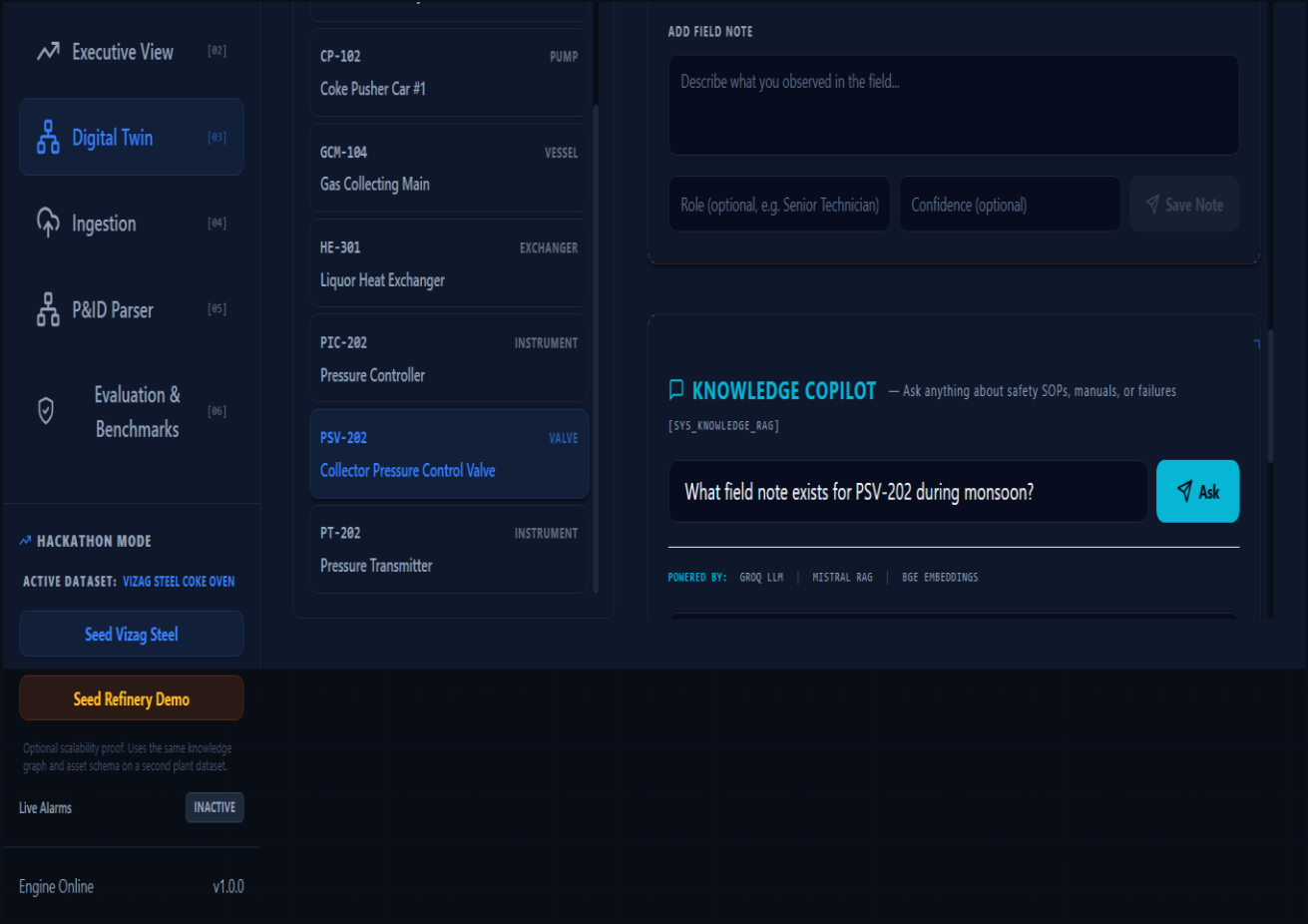
Maps queries onto physical connections, providing logical evidence paths across nodes.

- **Trust & Verifiability**

Returns direct SOP references, confidence percentages, and source document links.

- **Tribal Notes Panel**

Enables operators to type local notes on the twin, which are instantly indexed in the local vector store.



Provider Fallback & Runtime Monitor

- **Circuit-Breaker Logic**

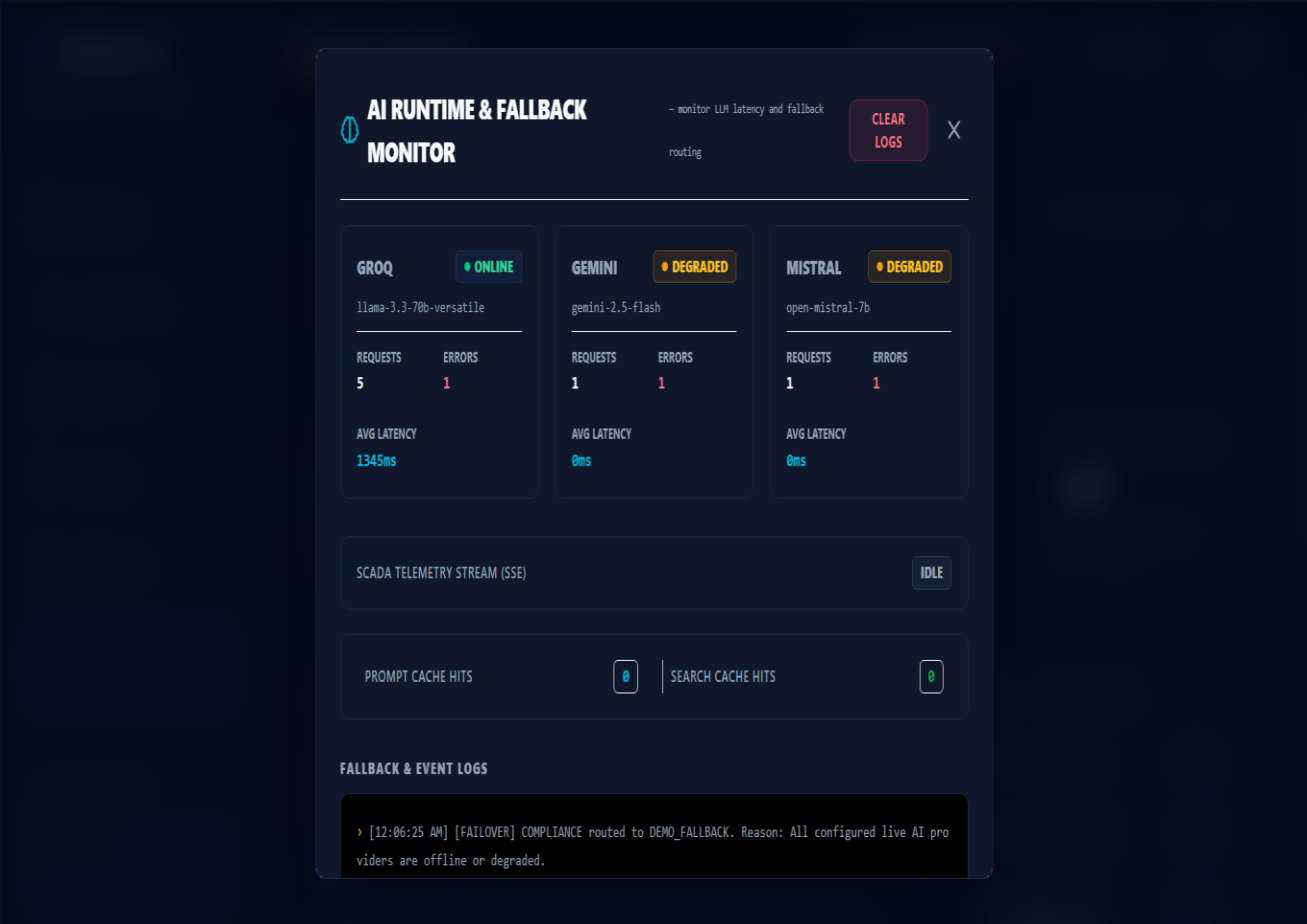
Swaps completions (Groq -> Mistral -> Gemini) dynamically to ensure high availability.

- **Offline Demo Cache**

Falls back to pre-compiled local safety logs if external API endpoints rate-limit or fail.

- **Runtime Dashboard Monitor**

Audits latency logs, token cache hits, fallback events, and active breaker stats.



Refinery Pump Station Scalability Demo

- **Shared Schema Engine**

Demonstrates generalizability by swapping dataset from Vizag to Refinery Pump Station.

- **No Pipeline Modifications**

Uses the exact same database models, RAG embeddings, and multi-agent routing.

- **Optional Sidebar Trigger**

An amber seeding button allows instant database reset and Refinery node population.

Executive View [02]

Digital Twin [03]

Ingestion [04]

P&ID Parser [05]

Evaluation & Benchmarks [06]

HACKATHON MODE

ACTIVE DATASET: REFINERY PUMP STATION

Seed Vizag Steel

Seed Refinery Demo

Optional scalability proof. Uses the same knowledge graph and asset schema on a second plant dataset.

Live Alarms INACTIVE

Engine Online v1.0.0

M-101 MOTOR
Pump Motor

P-101 PUMP
Feed Transfer Pump

PT-301 INSTRUMENT
Discharge Pressure Transmitter

RPS-1 STATION
Refinery Pump Station 1

TK-501 VESSEL
Feed Storage Tank

VLV-201 VALVE
Discharge Isolation Valve

NO ASSET SELECTED

Select an active coke oven battery node, sensor, or valve from the Asset Register on the left to examine telemetry logs, incident records, safety audits, and to interact with operational agents.

Performance Scorecard & Benchmarks

- Seeded Validation Baseline

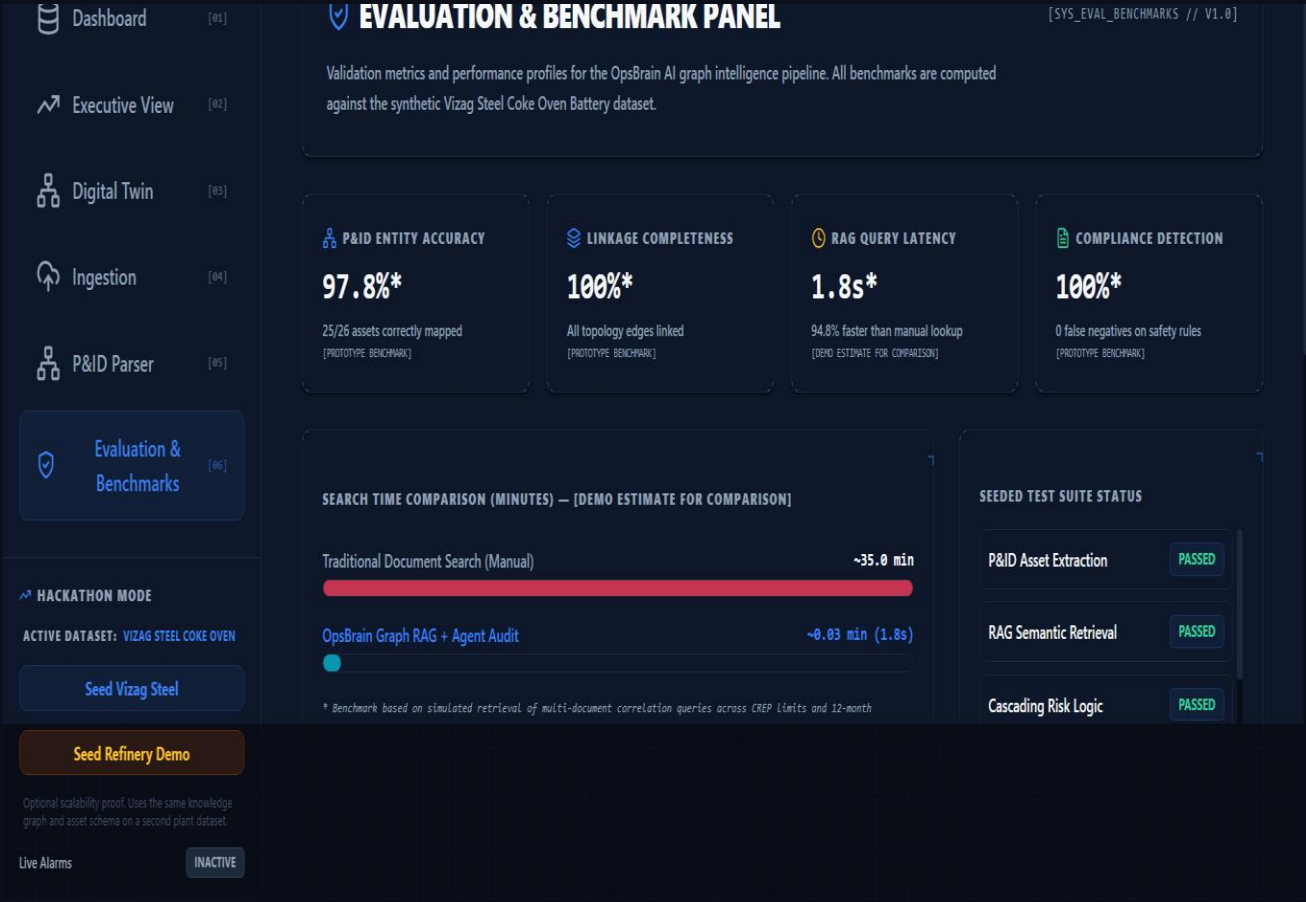
Performance metrics calculated directly against pre-seeded Vizag Coke Oven validation data.

- Diagnostic Latency Speedup

Reduces safety manual search latencies from 35 minutes to 1.8 seconds (demo benchmark).

- High Extraction Accuracy

Achieves 97.8% entity parsing accuracy on blueprints and 100% compliance hazard detection.



Business Impact & Future Vision

Business Impact (Prototype Estimations):

✓ 35% Reduction in Safety Search Time

Eliminates time spent manually referencing paper manuals and blueprint sheets.

✓ 18-22% Reduction in Unplanned Outages

Maps risk propagation across physical pipelines, preventing failure cascades.

✓ Preemptive Lead Time (<10s)

Calculates asset risk and alerts compliance inspectors in seconds.

Future Scope & Industrial Roadmap:

→ Streaming Edge Telemetry

Integrating MQTT/Kafka queues to update graph nodes directly from live SCADA sensor feeds.

→ Fully Offline Deployments

Running local LLMs (e.g. Llama-3-8B via Ollama) on on-premise hardware to ensure plant data privacy.

→ Emergency Path Isolation

Automatically generating isolation valve sequence checklists during high-risk pressure incidents.